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# Crystal Structure and Biochemical Characterization of Ketol-Acid Reductoisomerase from *Corynebacterium glutamicum*

Donghoon Lee,[†,‡,∥] Jiyeon Hong,[†,‡,∥] and Kyung-Jin Kim*[,†,‡,§]

[†]School of Life Sciences, BK21 Plus KNU Creative BioResearch Group, Kyungpook National University, Daehak-ro 80, Buk-ku, Daegu 702-701, Korea

[‡]KNU Institute for Microorganisms, Kyungpook National University, Daegu 41566, Republic of Korea

**ABSTRACT:** L-Valine belongs to the branched-chain amino acids (BCAAs) and is an essential amino acid that is crucial for all living organisms. L-Valine is industrially produced by the nonpathogenic bacterium *Corynebacterium glutamicum* and is synthesized by the BCAA biosynthetic pathway. Ketol-acid reductoisomerase (KARI) is the second enzyme in the BCAA pathway and catalyzes the conversion of (*S*)-2-acetolactate into (*R*)-2,3-dihydroxy-isovalerate, or the conversion of (*S*)-2-aceto-2-hydroxybutyrate into (*R*)-2,3-dihydroxy-3-methylvalerate. To elucidate the enzymatic properties of KARI from *C. glutamicum* (*Cg*KARI), we successfully produced *Cg*KARI protein and determined its crystal structure in complex with $NADP^+$ and two $Mg^{2+}$ ions. Based on the complex structure, docking simulations, and site-directed mutagenesis experiments, we revealed that *Cg*KARI belongs to Class I KARI and identified key residues involved in stabilization of the substrate, metal ions, and cofactor. Furthermore, we confirmed the difference in the binding of metal ions that depended on the conformational change.

**KEYWORDS:** *Corynebacterium glutamicum, ketol-acid reductoisomerase, L-valine*

## ■ INTRODUCTION

L-Valine is well-known as a branched-chain amino acid (BCAA) such as leucine and isoleucine and is one of the essential amino acids.[1] As L-valine can only be produced by microorganisms and plants, vertebrates are required to ingest L-valine from various additives or from food.[1−3] L-Valine is utilized in feed, food, dietary supplements, cosmetics, pharmaceuticals, and as a precursor of antibiotics or herbicides, as well as in many other commercial products.[4−6] Among them, feed additives for livestock account for the largest portion of industrial valine production, and in this form, L-valine is mainly consumed by poultry and pigs. Although the market size of L-valine is smaller than that of other amino acids, it is noted as an emerging industry in the essential amino acid market.[7,8] The global market of valine is growing at an average annual rate of more than 5%, and this trend is expected to continue for a while.

There are two methods to synthesize L-valine, namely, fermentation using microorganisms[9,10] and chemical synthesis.[11] In industry, most L-valine is produced by fermentation using *Corynebacterium glutamicum*.[12−14] *C. glutamicum* is a gram-positive, aerobic, and rod-shaped bacterium. It is widely utilized for biotechnological research and industrial production of various amino acids, because this strain is easy to handle, fast-growing, and capable of large-scale fermentation. Most importantly, *C. glutamicum* is a nonpathogenic bacterium that is harmless to humans.[15−18]

In *C. glutamicum*, L-valine is synthesized by the BCAA biosynthetic pathway, which consists of four steps (Figure 1A).[19,20] Ketol-acid reductoisomerase (KARI, EC 1.1.1.86; also known as acetohydroxy acid isomeroreductase) is the second enzyme in the BCAA pathway, producing the precursors of valine, leucine, and isoleucine (Figure 1A).[21] KARI functions as a bifunctional enzyme.[19,22] This enzyme catalyzes the conversion of (*S*)-2-acetolactate ((*S*)-2-AL) into (*R*)-2,3-dihydroxy-isovalerate, or the conversion of (*S*)-2-aceto-2-hydroxybutyrate into (*R*)-2,3-dihydroxy-3-methylvalerate (Figure 1A). These reactions are divided into two steps. First, electrons are transferred from the metal ions to migrate the alkyl group of the substrate, and then an NADPH-dependent reduction reaction occurs.[19,22] In addition, KARI enzymes can be divided into two classes, namely, Class I KARI and Class II KARI,[21,23] based on the amino acid sequence length and oligomeric status. Class I KARI contains around 340 amino acids and can catalyze as a dimeric form. On the other hand, the Class II KARI has a maximum of 490 amino acids and can function solely as a monomer due to an additional sequence.[21,23]

In the present study, we determined the crystal structure of KARI from *C. glutamicum* (*Cg*KARI) complexed with $NADP^+$ and two $Mg^{2+}$ ions and elucidated the cofactor and metal ions binding modes of the enzyme. In addition, we revealed the enzymatic properties of *Cg*KARI through kinetic analysis and site-directed mutagenesis experiments.

## ■ MATERIALS AND METHODS

**Preparation of *Cg*KARI Proteins.** The *Cg*KARI coding gene was amplified by the polymerase chain reaction (PCR) using genomic DNA from *C. glutamicum* strain ATCC 13032 as a template. The PCR product was then subcloned into the pET30a (Novagen) expression vector. The pET30a-*Cg*KARI was transformed into an *Escherichia coli* BL21 (DE3)-$T1^R$ strain, which was grown in 1 L of LB medium containing kanamycin (50 mg $L^{-1}$) at 37 °C to an OD600 of 0.6. After induction via the addition of 1.0 mM isopropyl 1-thio-*β*-D-

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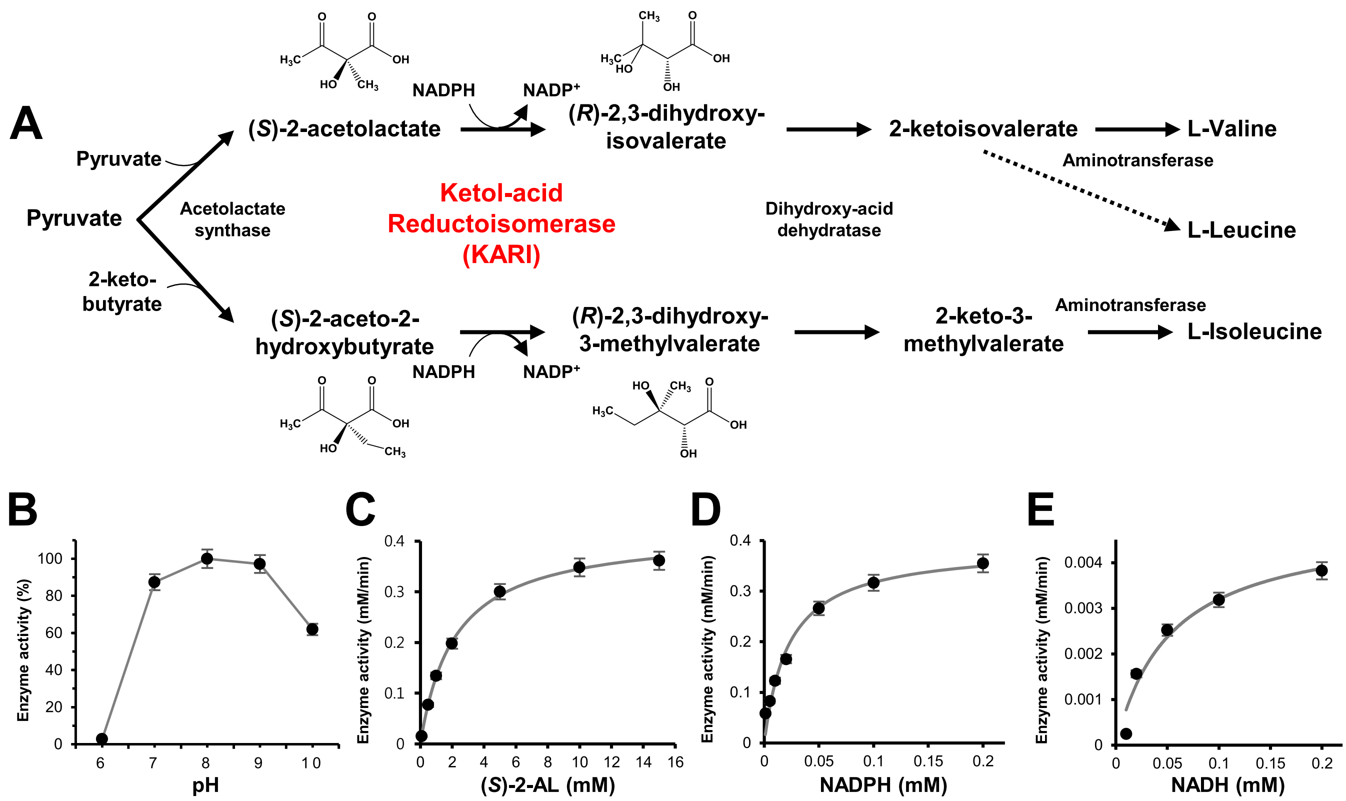


**Figure 1.** BCAA pathway and enzyme characterization of*Cg*KARI. The experiments were performed in triplicates. (A) BCAA pathway. (B) Optimal pH of *Cg*KARI. The effect of pH on *Cg*KARI activity was measured in pH range of 2 to 11. Relative activity was calculated based on the specific activity of *Cg*KARI at pH 8. (C) Kinetic curves of *Cg*KARI using various concentrations of (*S*)-2-AL. (D) Kinetic curves of *Cg*KARI using various concentrations of NADPH. (E) Kinetic curves of *Cg*KARI using various concentrations of NADH.

galactopyranoside (IPTG), the culture medium was further maintained for 20 h at 18 °C. After that, the culture medium was harvested by centrifugation at 4000*g* for 20 min at 4 °C. The cell pellet was resuspended in ice-cold buffer A (40 mM Tris-HCl, pH 8, and 150 mM NaCl) and disrupted by ultrasonication. The cell debris was removed through centrifugation at 13500*g* for 30 min at 4 °C, and the lysate was applied to a Ni-NTA agarose column (QIAGEN). After washing with buffer A containing 25 mM imidazole, the bound proteins were eluted with 300 mM imidazole in buffer A. Finally, a trace amount of contaminants was removed by size exclusion chromatography using a HiPrep 26/60 Sephacryl S-300 HR column (GE Healthcare Life Sciences) equilibrated with buffer A. All purification experiments were performed at 4 °C.

**Enzyme Kinetics of *Cg*KARI.** The activity of *Cg*KARI was measured using a decrease of NADPH at the wavelength of 340 nm. All assays were run at optimal pH and were performed with a 1 mL total volume of the reaction mixture. Also, reaction velocities were performed at room temperature for 2 min. The reaction mixture contained 0.1 M Tris-HCl, pH 8, 10 mM magnesium chloride, 1 mM 1,4-dithiothreitol (DTT), 30 $\mu$M *Cg*KARI protein, and various concentrations of (*S*)-2-AL (0.1 mM to 15 mM). The experiments were performed with (*S*)-2-acetolactic acid potassium salt from Santa Cruz Biotechnology. Cofactor specificity was measured at the reaction mixture of the same conditions, but the concentrations of (*S*)-2-AL were fixed at 10 mM. The concentration of NADPH and NADH was 0.001−0.2 mM and 0.01−0.2 mM, respectively. $V_{max}$ and $K_m$ values of substrates and cofactors were determined using the Michaelis−Menten equation, and results were plotted as a graph using OriginPro software (OriginLab Corporation, Massachusetts, USA).[24] The optimal pH was measured at the concentration exhibiting maximum activity (NADPH: 0.2 mM; (*S*)-2-AL: 10 mM), and four types of buffers were used for the measurement. The low pH values such as pH 2, 3, and 4 were measured with 0.1 M citrate buffer titrated with NaOH, and pH values of 5 and 6 were measured with 0.1 M Na citrate buffer titrated with HCl. The enzymatic activity in pH 7, 8, and 9 were analyzed by 0.1 M Tris-HCl buffer used in the purification process. The high pH values of 10 and 11 were analyzed using 0.1 M CAPS.

**Site-Directed Mutagenesis of *Cg*KARI.** Site-specific mutations were created using the QuikChange site-directed mutagenesis kit (Stratagene), and sequencing was performed to confirm the correct incorporation of the mutations. Mutant proteins were purified in the same manner as the wild-type, and activity of the mutant proteins was assayed under the above condition with 0.2 mM NADPH.

**Crystallization of *Cg*KARI.** Crystallization of the purified *Cg*KARI protein was initially performed with commercially available sparse-matrix screens including Index, PEG ion I and II (Hampton Research), Wizard Classic I and II, Wizard CRYO I and II (Rigaku Reagents), and Structure Screen I and II (Molecular Dimensions), using the hanging-drop vapor-diffusion method at 20 °C. Each experiment consisted of mixing 1.0 $\mu$L of protein solution (52 mg $mL^{-1}$ in 40 mM Tris-HCl, pH 8, and 150 mM NaCl) with 1.0 $\mu$L of reservoir solution and equilibrating the drop against 500 $\mu$L of reservoir solution. *Cg*KARI crystals were observed from several crystallization screening conditions. The best diffracting crystals appeared in day 2 using a reservoir solution consisting of 25% (w/v) polyethylene glycol (PEG) 3350, 0.1 M Bis-Tris, pH 5.5, and 0.2 M magnesium chloride at 20 °C.

**Data Collection and Structure Determination of *Cg*KARI.** The data were collected at the 7A beamline of the Pohang Accelerator Laboratory (PAL, Pohang, Korea) with a Quantum 270 CCD detector (ADSC, USA).[25] All data were then indexed, integrated, and scaled together using the HKL2000 software package.[26] The *Cg*KARI crystals in complex with $NADP^+$ belonged to $P2_12_12_1$ with unit cell parameters $a$ = 84.9 Å, $b$ = 90.1 Å, $c$ = 157.8 Å, $\alpha = \beta = \gamma$ = 90°. Assuming four molecules of *Cg*KARI per asymmetric unit, the crystal volume per unit of protein mass was 2.09 $Å^3$ $Da^{1-}$, which corresponds to a solvent content of approximately 41.27%.[27] The structures of



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*Cg*KARI were determined by molecular replacement with the CCP4 version of MOLREP[28] using the structure of KARI from *Mycobacterium tuberculosis* (PDB code 4YPO) as a search model. The model building was performed manually using the program WinCoot,[29] and refinement was performed with CCP4 refmac5.[30] The data statistics are summarized in Table 2, and the refined *Cg*KARI structure in complex with $NADP^+$ was deposited in the protein data bank under the PDB code 6JX2.

**Size-Exclusion Chromatographic Analysis.** To identify the oligomeric status of *Cg*KARI, analytical size-exclusion chromatography was performed using a Superdex increase 200 10/300 GL column (GE Healthcare Life Sciences) equilibrated with 40 mM Tris-HCl, pH 8, and 150 mM NaCl. A protein sample of 1 mL with a concentration of 1 mg $mL^{-1}$ was analyzed. The molecular weights of the eluted samples were calculated based on the calibration curve drawn using standard samples of ferrin (440 kDa), conalbumin (75 kDa), carbonic anhydrase (29 kDa), and ribonuclease A (13.7 kDa) (GE Healthcare Life Sciences).

**Molecular Docking Simulation.** A molecular docking simulation of (*S*)-2-AL to *Cg*KARI structure was performed by AutoDock Vina software.[31] A *Cg*KARI structure in complex with $NADP^+$ cofactor (PDB code of 6JX2) and the ligand molecule of *Cg*KARI, (*S*)-2-AL, were prepared using the JLigand software. For the docking simulation, the *pdbqt* files were generated using the AutoDock Vina manual. The grid size for (*S*)-2-AL was $x = 22$, $y = 16$, and $z = 20$, and the grid center was designated at $x = 41.446$, $y = 36.571$, and $z = 26.653$. The final conformations produced in this simulation were checked using PyMOL software.[32]

**Metal Activity Assay.** The metal activity was measured at the concentration exhibiting maximum activity (NADPH: 0.2 mM; (*S*)-2-AL: 10 mM), and the composition of the other reaction mixtures proceeded under the same conditions as the enzyme kinetic assay. Six types of metal ions were used for the metal activity assay, including magnesium chloride ($MgCl_2$), cobalt chloride ($CoCl_2$), nickel chloride ($NiCl_2$), copper chloride ($CuCl_2$), zinc chloride ($ZnCl_2$), and manganese chloride (MnCl2). Metal ions were measured at various concentrations by diluting 1 M stock.

## RESULTS AND DISCUSSION

**Production and Enzymatic Properties of *Cg*KARI.** To characterize the biochemical properties of KARI in *C. glutamicum*, the gene coding for KARI from the *C. glutamicum* ATCC 13032 strain was cloned into the recombinant expression vector pET30a and successfully overexpressed in the *Escherichia coli* BL21 (DE3)-$T1^R$ strain. The recombinant KARI protein was purified to homogeneity by a two-step procedure comprising affinity chromatography and size-exclusion chromatography. The molecular weight of the purified *Cg*KARI enzyme was confirmed by SDS-PAGE to be 36.1 kDa. Based on the results, we identified that *Cg*KARI belongs to Class I KARI. The yield of the purified *Cg*KARI was 100 mg protein $L^{-1}$ of the cell culture.

First, to investigate the optimal pH of *Cg*KARI, we measured the KARI activity of *Cg*KARI at various pH ranging from 2 to 11. In the pH range 2−6 and at pH 11, we could not measure the KARI activity, due to heavy precipitation of the protein. Thus, we measured the enzyme activity from pH 7 to 10. *Cg*KARI showed the highest enzyme activity at pH 8, and the relative activity at pH 7, 9, and 10 was 87, 97, and 61%, respectively (Figure 1B). These results are similar to the reported pH range of KARI from *Escherichia coli* and *Oryza sativa*,[33] and most KARIs except for some specific strains, such as thermoacidophilic archeae,[34] are expected to show activity between pH 7 and 10.[35]

At the optimal pH, we performed the enzyme kinetics analysis of the *Cg*KARI by using (*S*)-2-AL as the substrate. Reaction rates corresponding to various concentrations of (*S*)-2-AL were plotted and determined to obey Michaelis−Menten kinetics. The $k_{cat}$ and $K_m$ values of (*S*)-2-AL were 14.089 $min^{-1}$ and 2.161 mM, respectively (Figure 1C and Table 1). Based on these results, the $k_{cat}/K_m$ value for the (*S*)-2-AL substrate was calculated as 6.519 $mM^{-1}$ $min^{-1}$ (Table 1). Furthermore, we also performed the enzyme kinetics analysis of the cofactor. According to the analysis results, *Cg*KARI prefers NADPH as a

**Table 1. Kinetic Analysis of *Cg*KARI**

| | $V_{max}$ [mM $min^{-1}$] | $K_m$ [mM] | $k_{cat}$ [$min^{-1}$] | $k_{cat}/K_m$ [$mM^{-1}$ $min^{-1}$] |
|---|---|---|---|---|
| (*S*)-2-acetolactate | 0.41967 | 2.16103 | 14.08963 | 6.51986 |
| NADPH | 0.38892 | 0.02258 | 13.05725 | 578.2663 |
| NADH | 0.00491 | 0.05353 | 0.16484 | 1.79716 |

**Table 2. Data Collection and Structural Refinement Statistics**

| | *Cg*KARI_$NADP^+$ |
|---|---|
| PDB code | 6JX2 |
| **Data Collection** | |
| wavelength (Å) | 0.97934 |
| space group | $P2_12_12_1$ |
| cell dimensions | |
| *a*, *b*, *c* (Å) | 84.9, 90.1, 157.8 |
| $\alpha$, $\beta$, $\gamma$ (deg) | 90.00, 90.00, 90.00 |
| resolution range (Å) | 78.9−2.6 |
| highest resolution shell (Å) | 2.64−2.60 |
| unique reflections | 37991 |
| $R_{sym}$ or $R_{merge}$ | 9.4 (30.5) |
| $R_{meas}$ (%) | 10.6 (35.6) |
| $I/\sigma$ (*I*) | 20.6 (3.15) |
| completeness (%) | 98.1 (97.3) |
| redundancy | 4.2 (3.4) |
| **Refinement** | |
| resolution (Å) | 78.9−2.6 |
| no. reflections | 35327 |
| $R_{work}/R_{free}$ | 17.9 (26.6) |
| no. atoms | 9995 |
| protein | 9754 |
| ligand/ion | 143 |
| water | 98 |
| *B*-factors | 54.779 |
| protein | 54.546 |
| ligand/ion | 64.11 |
| water | 44.988 |
| **RMS Deviations** | |
| bond lengths (Å) | 0.011 |
| bond angles (deg) | 1.548 |
| planarity (Å) | 0.006 |
| chirality ($Å^3$) | 0.081 |
| **Ramachandran Plot** | |
| favored (%) | 95.74 |
| allowed (%) | 4.19 |
| outliers (%) | 0.08 |

[a]The numbers in parentheses are statistics from the highest resolution shell. [b]$R_{sym} = \Sigma |I_{obs} - I_{avg}|/I_{obs}$, where $I_{obs}$ is the observed intensity of individual reflection and $I_{avg}$ is the average over symmetry equivalents. [c]$R_{work} = \Sigma ||F_o| - |F_c||/\Sigma |F_o|$, where $|F_o|$ and $|F_c|$ are the observed and calculated structure factor amplitudes, respectively. $R_{free}$ was calculated with 5% of the data.


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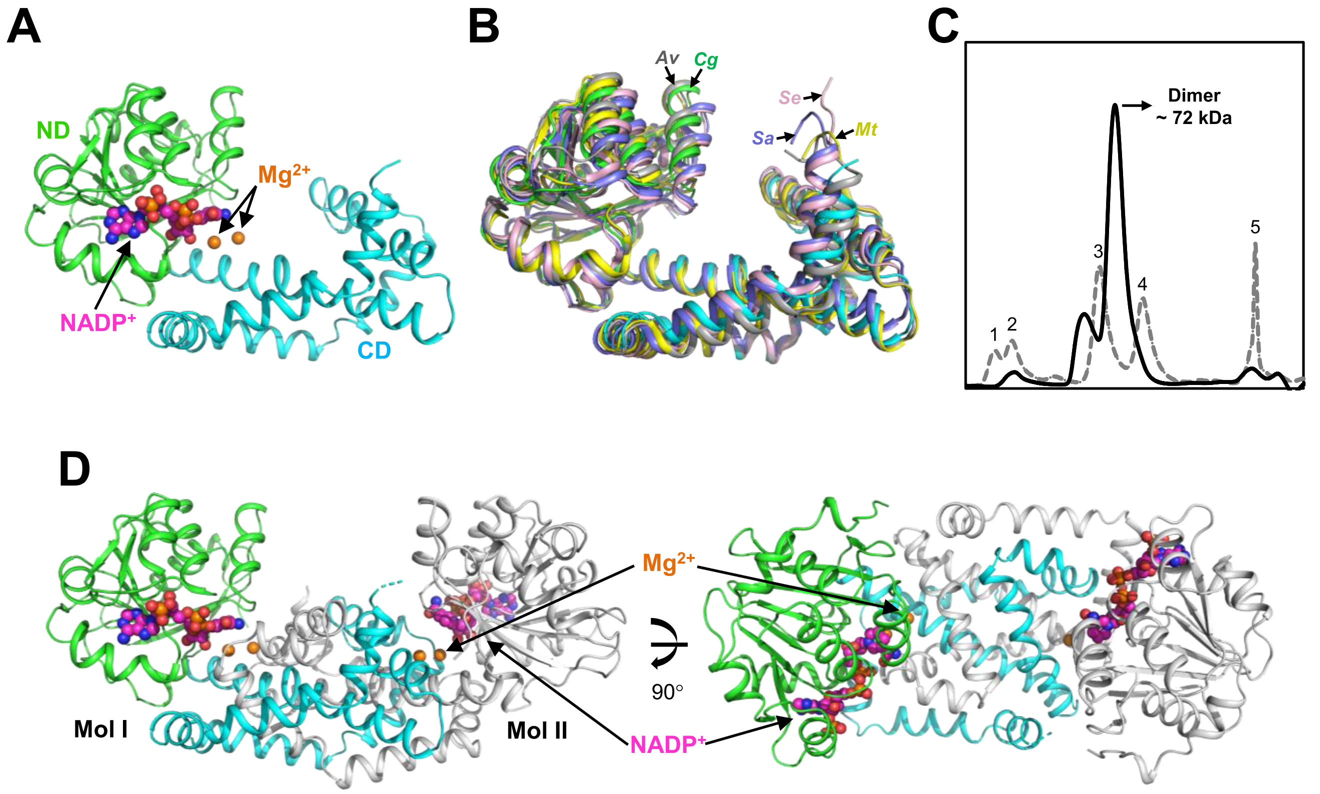


**Figure 2.** Overall structure of *Cg*KARI. (A) The monomeric structure of *Cg*KARI. The *Cg*KARI structure is shown as a cartoon diagram. The ND and CD are distinguished by different colors of green and cyan, respectively, and are labeled appropriately. The bound $NADP^+$ and $Mg^{2+}$ are shown as magenta and orange colored spheres, respectively. (B) Superposition of the monomeric structures. The monomeric structure of *Cg*KARI, *Mt*KARI, *Av*KARI, *Se*KARI, and *Sa*KARI is shown as a cartoon diagram. *Cg*KARI is presented with a color scheme the same as in (A). *Mt*KARI, *Av*KARI, *Se*KARI, and *Sa*KARI are presented with colors of yellow, gray, light pink, and light blue, respectively, and are labeled appropriately. (C) Size-exclusion chromatography of *Cg*KARI. The *Cg*KARI and standard samples are distinguished with solid and dotted lines, respectively. The numbers 1, 2, 3, 4, and 5 indicate void and standard samples of ferritin (440 kDa), conalbumin (75 kDa), carbonic anhydrase (29 kDa), and ribonuclease A (13.7 kDa), respectively. (D) Dimeric structure of *Cg*KARI. The dimeric structure of *Cg*KARI is presented as a cartoon diagram. Mol I is presented with a color scheme the same as in (A). Mol II is shown with a gray color. The bound $NADP^+$ and $Mg^{2+}$ are shown as sphere models with colors of magenta and orange, respectively. The right figure is the left figure rotated horizontally by 90°.

cofactor rather than NADH and the $k_{cat}$ and $K_m$ value of NADPH were 13.057 $min^{-1}$ and 0.022 mM, respectively (Figure 1D, 1E, and Table 1). On the basis of the enzyme kinetics analysis, we confirmed that *Cg*KARI utilizes NADPH as a cofactor in the catalysis.

**Overall Structure of *Cg*KARI.** To elucidate the molecular mechanism and enzymatic properties of *Cg*KARI, we determined its crystal structure at 2.6 Å resolution. The refined structure was in good agreement with the X-ray crystallographic statistics for bond angles, bond lengths, and other geometric parameters (Table 2). The overall structure of *Cg*KARI is similar to that of Class I KARIs (Figure 2A and 2B). The DALI server[36] showed that the *Cg*KARI structure is highly homologous to Class I KARIs from *Mycobacterium tuberculosis* (*Mt*KARI, PDB code 4YPO and Z-score 43.4),[37] *Azotobacter vinelandii* (*Av*KARI, PDB code 4XIY and Z-score 42.6),[23] *Slackia exigua* (*Se*KARI, PDB code 4KQW and Z-score 41.6),[38] and *Staphylococcus aureus* (*Sa*KARI, PDB code 5W3K and Z-score 40.9).[39] The monomeric structure of *Cg*KARI consists of 17 $\alpha$-helices ($\alpha$1−$\alpha$17) and eight $\beta$-strands ($\beta$1−$\beta$8) (Figure 2A) and is divided into two distinct domains, the N-terminal domain (ND) and the C-terminal domain (CD). The ND (Met1−Thr184) consists of nine $\alpha$-helices and eight $\beta$-strands and shows the Rossmann fold that is well-known as an NADPH cofactor binding fold (Figure 2A). The CD (Phe185−Ala338) consists of only eight $\alpha$-helices and is involved in active site formation (Figure 2A). The active site is located in the cleft of the ND and CD, and the active site pocket is completed by the CD of the neighboring monomer (Figure 2D).

The KARI enzymes are known to have a various oligomeric status, such as dimer, tetramer, and dodecamer.[34,35] Among them, *Cg*KARI functions as a dimer. The four molecules in the asymmetric unit can be assigned to two dimers, and the size-exclusion chromatography confirmed that *Cg*KARI exists as a dimer in solution (Figure 2C). Dimerization of *Cg*KARI is mainly mediated by the CD of the two monomers, and the CD of molecule I (Mol I) is interdigitated with the CD of molecule II (Mol II), which enables a tightly bound dimer (Figure 2D). All $\alpha$-helices ($\alpha$10−$\alpha$17) of the CD are directly involved in the stabilization of the dimer through 37 hydrogen bonds and 32 salt bridges. Through the PISA software,[40] it is calculated that a total solvent-accessible interface area of 13254 Å$^2$ is buried upon dimerization.

**Active Site of *Cg*KARI.** Although we did not add $NADP^+$ molecules during the protein production and crystallization processes, we fortunately observed the electron density for the $NADP^+$ molecule in our structure (Figure 3A). The active site of *Cg*KARI is formed in the cleft of the ND and CD (Figure 3B). The CD of the neighboring monomer also contributes to


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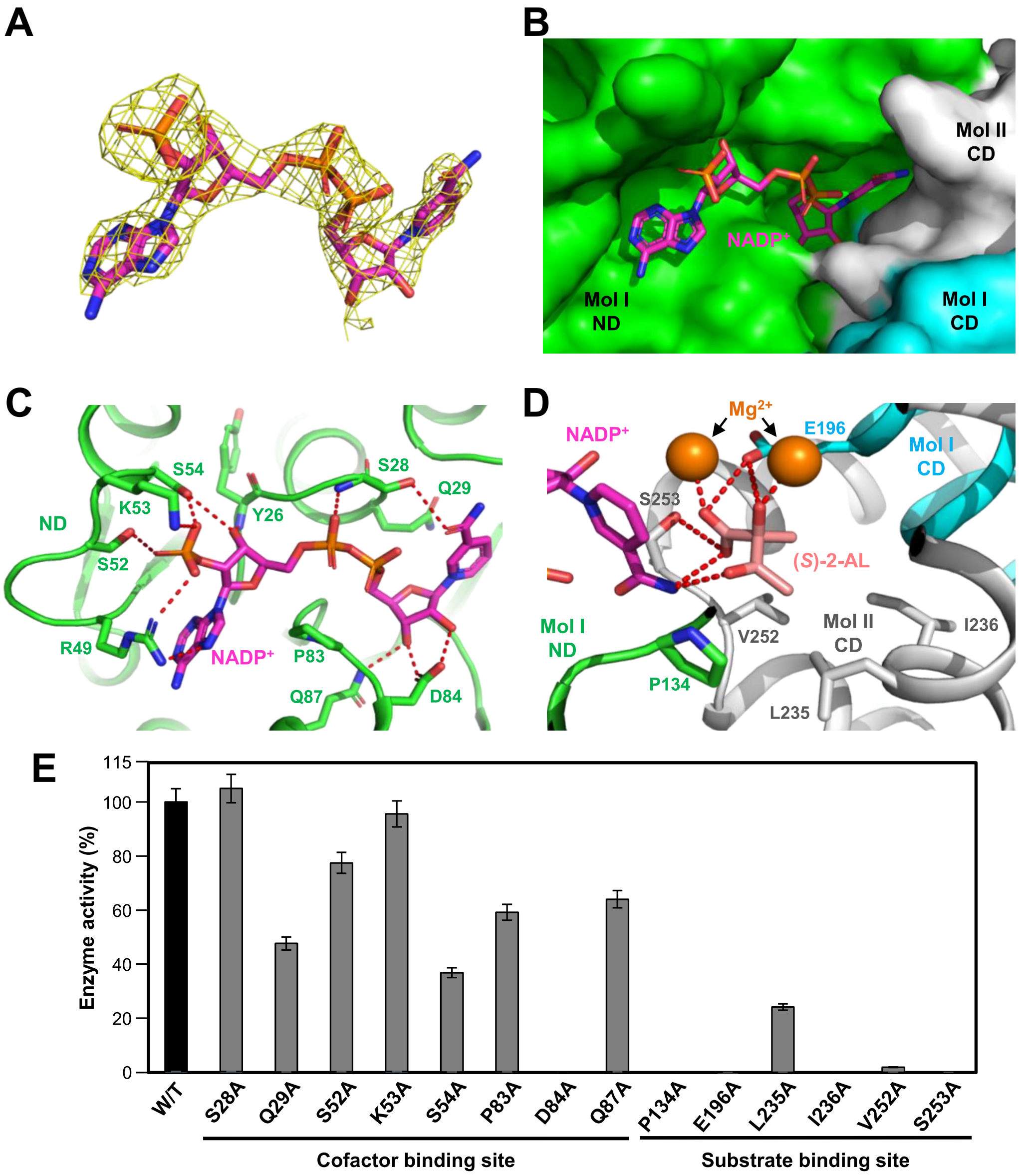


**Figure 3.** Cofactor and substrate binding mode of *Cg*KARI. (A) Electron density map of NADP$^+$. The $2F_o-F_c$ electron density map of the bound NADP$^+$ is shown as a yellow colored mesh and contoured at $1\sigma$. The bound NADP$^+$ is presented with a stick model with a magenta color. (B) A surface model of the active site of *Cg*KARI. The *Cg*KARI structure is shown as a surface model. Three domains are distinguished with different colors of green, cyan, and gray for Mol I ND, Mol I CD, and Mol II CD, respectively, and are labeled appropriately. The bound NADP$^+$ molecule is shown as a stick model with a magenta color. (C) NADP$^+$ binding mode of *Cg*KARI. The *Cg*KARI structure is presented with a cartoon diagram with the same color scheme as in (B). The residues involved in the NADP$^+$ binding are shown as a stick model and are labeled appropriately. The hydrogen bond involved in the cofactor binding are shown as red-colored dotted lines. The bound NADP$^+$ molecule is shown as a stick model with a magenta color. (D) (*S*)-2-AL binding mode of *Cg*KARI. The *Cg*KARI structure is presented with a cartoon diagram with the same color scheme as in (B). The residues involved in the stabilization of substrate are shown as a stick model and are labeled appropriately. The hydrogen bond involved in the substrate binding are shown as red-colored dotted lines. The bound molecules, NADP$^+$, (*S*)-2-AL, and $Mg^{2+}$, are presented with stick models and spheres with colors of magenta, salmon, and orange, respectively. (E) Site-directed mutagenesis of *Cg*KARI. Residues involved in stabilization of substrate and cofactor are replaced by alanine residues. The relative activity of recombinant mutant proteins was measured and compared with that of wild-type *Cg*KARI. Each experiment was performed in triplicate.

the formation of the active site. When *Cg*KARI dimerizes, the complete active site pocket is formed, so *Cg*KARI can exert its catalytic activity (Figure 3B). The residues that constitute the active site are highly conserved in the KARI enzymes, and these conserved residues are directly involved in stabilizing the cofactor and substrate. The NADP$^+$ cofactor is mainly stabilized by the ND. One $\alpha$-helix ($\alpha$6) and five $\beta-\alpha$ connecting loops ($\beta$1-$\alpha$3, $\beta$2-$\alpha$4, $\beta$4-$\alpha$6, $\beta$5-$\alpha$7, and $\beta$6-$\alpha$8) of the ND form the cofactor binding site. The nicotinamide riboside moiety of NADP$^+$ is buried in the active site pocket while the adenosine moiety of NADP$^+$ is exposed on the surface of the ND (Figure 3B). The phosphate moiety of adenosine is mainly stabilized by direct hydrogen bonding with the side chains of Arg49, Ser52, Lys53, and Ser54 (Figure 3C). The Arg49 is also hydrogen bonded with the adenine ring of NADP$^+$, and Ser54 further interacts with ribose ring moiety of adenosine through the hydrogen bond. The loop including these residues is highly conserved in the KARI enzymes as a



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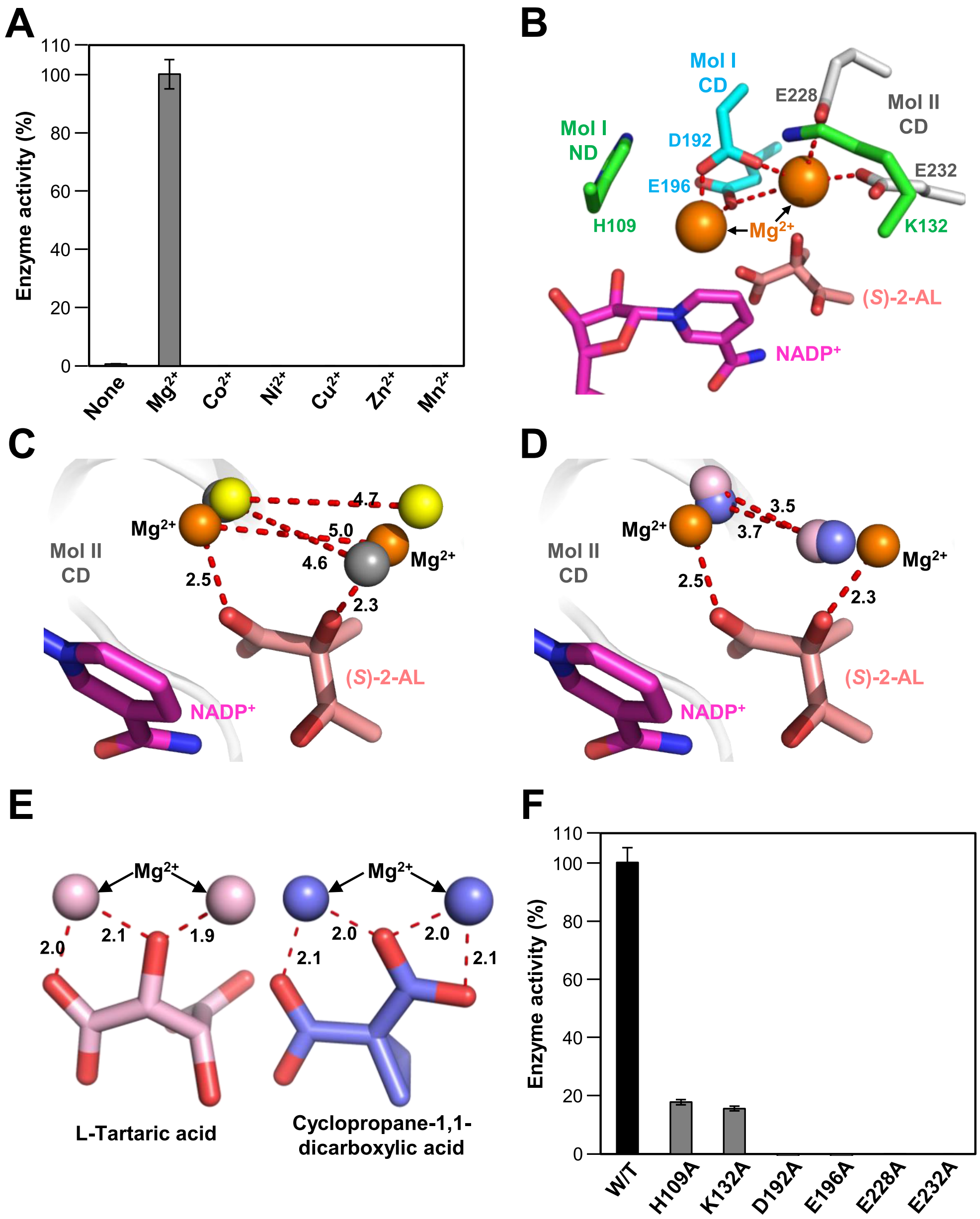


**Figure 4.** Metal ions binding mode of *Cg*KARI. (A) *Cg*KARI activity depending on metal ions. Enzyme activity of *Cg*KARI was measured using the six metal ions and was also measured without metal ions. Each experiment was performed in triplicate. (B) Metal ions binding mode of *Cg*KARI. The conserved residues of *Cg*KARI are shown as a stick model with the same color scheme as in Figure 3B and labeled appropriately. NADP$^+$, (*S*)-2-AL, and $Mg^{2+}$ ions are presented with stick models and spheres with colors of magenta, salmon, and orange, respectively. The interactions between metal ions and conserved residues are shown as red-colored dotted lines. (C) Metal ions binding mode of KARI enzymes without conformational change. Metal ions of *Cg*KARI, *Mt*KARI, and *Av*KARI are superimposed and are shown as spheres with colors of orange, yellow, and gray, respectively. The cofactor and substrate of *Cg*KARI are presented with a stick models with the same color scheme as in (B). The distance between the metal ions is shown as red-colored dotted lines. (D) Metal ions binding mode of KARI enzymes with conformational change. Metal ions of *Cg*KARI, *Se*KARI, and *Sa*KARI are superimposed and are shown as spheres with colors of orange, light pink, and light blue, respectively. The cofactor and substrate of *Cg*KARI are presented with a stick models with the same color scheme as in (B). The distance between the metal ions is shown as red-colored dotted lines. (E) Interaction between metal ions and substrate analog. The metal ions and substrate analog of *Se*KARI and *Sa*KARI are shown as spheres and stick models with colors of light pink and light blue. The distance between the metal ions and substrate analog is shown as red-colored dotted lines. (F) Site-directed mutagenesis of *Cg*KARI. Conserved residues involved in stabilization of metal ions are replaced by alanine residues. The relative activity of recombinant mutant proteins was measured and compared with that of wild-type *Cg*KARI. Each experiment was performed in triplicate.

$\beta2\alpha$B loop and directly influences the cofactor specificity of KARI.[38] In addition, the main chain of Tyr26 contributes to the stabilization of the ribose ring moiety of adenosine (Figure 3C). A pyrophosphate moiety is located at the entrance of the active site pocket and forms a hydrogen bond with the main chain nitrogen atom of Ser28 (Figure 3C). Pro83 does not directly interact with the cofactor but might contribute to forming the inlet of active site pocket and assist NADP$^+$ in locating the proper binding site. The side chains of Ser28, Gln29, Asp84, and Gln87 directly interact with nicotinamide and ribose through hydrogen bonds and stabilize the


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nicotinamide riboside moiety buried in the pocket (Figure 3C).

To elucidate the substrate binding mode of *Cg*KARI, we performed molecular docking calculations of (*S*)-2-AL into the *Cg*KARI structure. In the molecular docking simulations, (*S*)-2-AL fits well into the negatively charged pocket of the active site of *Cg*KARI and the carboxyl moiety of the substrate is stabilized by direct hydrogen bonding with the side chains of Glu196 and Ser253 (Figure 3D). The active site pocket is lined internally with hydrophobic residues, such as Pro134, Leu235, Ile236, and Val252, and these residues accommodate the methyl groups of the C2 and C3 of (*S*)-2-AL (Figure 3D). In addition, those four hydrophobic residues stabilize the catalytic intermediate when the alkyl group migrates from C2 to C3 (Figure 3D).

To identify the residues involved in stabilization of the cofactor and substrate, we performed site-directed mutagenesis experiments and compared the enzyme activity with that of wild-type *Cg*KARI. When the residues were replaced with alanine, Q29A, S52A, K53A, S54A, P83A, Q87A, and L235A mutants were less active than the wild-type *Cg*KARI (Figure 3E). In addition, D84A, P134A, E196A, I236A, V252A, and S253A mutants completely lost the KARI activity (Figure 3E). Unlike the other mutants, S28A exhibited a slightly increased activity (Figure 3E), indicating that the residue does not seem to be crucial for binding of the cofactor. As a result, we confirmed that Ser28, Gln29, Ser52, Lys53, Ser54, Pro83, Asp84, Gln87, Pro134, Glu196, Leu235, Ile236, Val252, and Ser253 residues are mainly involved in the cofactor and substrate binding.

**Metal Binding Mode of *Cg*KARI.** It is well-known that the KARI enzyme utilizes two metal ions during the catalytic reaction. The two metal ions do not simply aid the catalytic mechanism but instead participate directly in the process and initiate the catalytic reaction.[20,41] Previous studies have reported that $Mg^{2+}$ is specifically required for the alkyl migration reaction, whereas other metal ions can be utilized for the NADPH-dependent reduction reaction.[21,22,33] To confirm that *Cg*KARI uses metal ions for its catalysis, we measured the enzyme activity without metal ions, and no enzyme activity is observed (Figure 4A). We then measured the enzyme activity using six metal ions, $Mg^{2+}$, $Co^{2+}$, $Ni^{2+}$, $Cu^{2+}$, $Zn^{2+}$, and $Mn^{2+}$. Interestingly, we only observed the KARI activity by addition of the $Mg^{2+}$ ion (Figure 4A). On the other hand, *Cg*KARI had no enzyme activity for other metal ions (Figure 4A). Moreover, we observed heavy precipitation of *Cg*KARI on the addition of the other metal ions to the reaction mixture. These observations indicate that *Cg*KARI utilizes the $Mg^{2+}$ ion for two-step enzyme catalysis and that the metal ion is also crucial for protein stability.

In the structure of *Cg*KARI, two $Mg^{2+}$ ions were tightly bound in the active site pocket. We expect that the crystallization buffer contains $Mg^{2+}$ and that it might influence the tight crystal packing of the *Cg*KARI crystal. Two $Mg^{2+}$ ions are surrounded by six highly conserved residues, His109, Lys132, Asp192, Glu196, Glu228, and Glu232, and the interaction between metal ions and conserved residues is involved in both the stabilization of the metal ions and the enzyme catalysis (Figure 4B). As a conformational change of ND upon substrate binding was not observed in the structure of *Cg*KARI, the detailed investigation of the metal binding mode was performed by the structural comparisons with other KARI enzymes. First, the distance between two metal ions is somewhat different between KARI enzymes depending on the conformational change. In the KARI enzymes without conformation change, including *Cg*KARI, *Mt*KARI, and *Av*KARI, the distances between the metal ions were 4.6, 4.7, and 5.0 Å, respectively (Figure 4C). On the other hand, the KARI enzymes in which substrate analogs were bound, such as *Se*KARI and *Sa*KARI, showed shorter distances between metal ions of 3.5 and 3.7 Å, respectively (Figure 4D). In addition, the interactions between metal ions and substrate analogs are different. The (*S*)-2-AL of *Cg*KARI is about 2.5 Å apart from $Mg^{2+}$, whereas the substrate analogs in *Se*KARI and *Sa*KARI are located at about 2.0 Å distance, and more tight interactions between metal ions and substrate analogs were observed (Figure 4E).[38,39] Based on these results, the conformational change caused by the binding of substrate influences the position of the metal ion, and the movement of the metal ion enables a stronger interaction with the substrate, resulting in a catalytic reaction.

To confirm the residues involved in the enzyme catalysis, we performed site-directed mutagenesis experiments and compared the KARI activity with that of the wild-type enzyme. When the residues involved in the stabilization of metal ions were replaced with alanine, the positively charged residues such as His109 and Lys132 retained slight enzyme activity, whereas the negatively charged residues, Asp192, Glu196, Glu228, and Glu232, showed complete loss of activity (Figure 4F). These results suggest that although all six conserved residues are involved in metal stabilization, the four negatively charged residues are directly involved in the enzyme catalysis.

In summary, we successfully produced *Cg*KARI protein and demonstrated the enzymatic properties through various experiments. We determined the crystal structure of *Cg*KARI complexed with $NADP^{+}$ and elucidated the cofactor binding mode of *Cg*KARI. Through the molecular docking simulations of (*S*)-2-AL, we proposed the substrate binding site and its binding mode. Furthermore, we confirmed that *Cg*KARI utilizes $Mg^{2+}$ in the enzyme catalysis, and we elucidated the crucial residues for metal binding and enzyme catalysis. Finally, we also confirmed the effect of conformational change on metal ions and predicted how metal ions interact with the substrate during catalysis. Our findings might provide valuable information to develop highly efficient valine production. Because *Cg*KARI is an enzyme involved in production of leucine and isoleucine, the study might also contribute to further improvement of productivity of these amino acids.

## ■ AUTHOR INFORMATION

### Corresponding Author

*E-mail: kkim@knu.ac.kr. Tel.: +82-53-950-5377. Fax: +82-53-955-5522.

### ORCID

Kyung-Jin Kim: 0000-0002-0375-2561

### Present Address

[§]Kyung-Jin Kim, Ph.D., Structural and Molecular Biology Laboratory, School of Life Sciences, Kyungpook National University, Daehak-ro 80, Buk-ku, Daegu 702-701, Korea.

### Author Contributions

[‖]D.L. and J.H. contributed equally to this work.

### Notes

The authors declare no competing financial interest.



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